Shivansh Bhardwaj

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Summary

Computer Science graduate (VIT Bhopal) with an IIT Guwahati credential in Artificial Intelligence and Machine Learning (Computer Vision, NLP, Transformers, Reinforcement Learning, Spark, MLOps). Building computer vision, LLM/RAG and multimodal Generative AI projects with measured, reproducible results. Seeking an entry-level AI/ML Engineer role.

Education

Indian Institute of Technology (IIT) Guwahati — Guwahati, India

Credit-Linked Certification Program in Artificial Intelligence & Machine Learning (CGPA: 6.34/10) *Certificate expected Oct 2026*

- **Coursework:** Computer Vision & Object Detection (YOLO), NLP & Transformer Architectures, Reinforcement Learning, Distributed Computing (Spark), MLOps Fundamentals

Vellore Institute of Technology (VIT), Bhopal — Bhopal, India

B.Tech in Computer Science (Core) (CGPA: 7.07/10) *Sep 2021 – Sep 2025*

Technical Skills

Languages: Python, SQL, Java
Computer Vision: PyTorch, OpenCV, YOLO (Ultralytics), torchvision, Roboflow
NLP / GenAI: Generative AI, LLMs & VLMs, RAG, Transformers, Hugging Face, LangChain, Embeddings & Vector Search (FAISS), Hybrid Search & Re-ranking, Prompt Engineering
Machine Learning: Deep Learning, Fine-tuning, Scikit-learn, XGBoost, NumPy, Pandas, Reinforcement Learning, Apache Spark (PySpark)
MLOps: Docker, FastAPI, GitHub Actions, MLflow, AWS (S3, EC2), pytest
Web: HTML, CSS, JavaScript
Core CS: Data Structures & Algorithms, OOP, DBMS, Git

Projects

Real-Time Object Detection API | *PyTorch, Ultralytics YOLO26, FastAPI, Docker, pytest* *GitHub | Sep 2026*

- Fine-tuned a YOLO26n detector on the public African Wildlife dataset, reaching mAP50 0.975 and mAP50-95 0.847 on the 227-image held-out test split.
- Served the model through a FastAPI REST endpoint that returns bounding boxes, class names and confidence scores as JSON, with a 50 ms median server inference time on an RTX 3050 Ti GPU (100-request benchmark).
- Packaged the service in a Docker image (CPU-only PyTorch, non-root user) and covered it with a pytest test suite.

Multimodal Document RAG System | *Qwen2.5-VL, Hugging Face Transformers, OCR, BM25, FAISS* *GitHub | Sep 2026*

- Built a document question-answering pipeline that OCRs and indexes scanned pages, retrieves the relevant page and has Qwen2.5-VL-3B (4-bit) read the answer from the page image on a 4 GB laptop GPU.
- Compared BM25, dense and hybrid retrieval on 120 questions over 45 documents; BM25 ranked the correct scanned page first for 96.7% of questions.
- Reached 96.7% end-to-end answer accuracy (116/120), with every remaining error traced to retrieval rather than the vision-language reader.

RAG Retrieval Benchmark & CI Gate | *BM25, Sentence-Transformers, FAISS, FastAPI, GitHub Actions GitHub | Oct 2026*

- Benchmarked BM25, dense (bge-small), hybrid Reciprocal Rank Fusion and cross-encoder re-ranking on BEIR SciFact (300 test queries, 5,183 documents) with paired significance tests.
- Hybrid retrieval reached nDCG@10 0.727 and Recall@100 0.965 versus 0.676 for BM25 ($p < 0.001$); documented that cross-encoder re-ranking did not improve on hybrid for this dataset.
- Shipped a FastAPI search service and a GitHub Actions regression gate that fails the build if nDCG@10 drops more than 0.02 below the committed baseline.

Experience

BlueEra Softech — Jan 2025 – Mar 2025

Web Development Intern

- Completed a 2-month web development internship working with HTML, CSS, JavaScript and Python.

Achievements

- Winner, Football – Aahvaan '23 Sports Tournament, VIT Bhopal University (May–Aug 2023)
- Silver, Inter-School Table Tennis Tournament; represented at SGFI State Table Tennis
- 3rd place, 100m Freestyle Swimming – Khel Mahakumbh, Vadodara (Govt. of Gujarat), 2015-16
- Participated, 64th District-Level School Football (U-19), Sikar, 2019-20